Group Theory

Group Theory Worked Solutions

Problem (August '05, Problem 2): Let G be a finitely generated group.

(a) If H is a finite group, show that there exist only finitely many group homomorphisms from G to H.

(b) If n is a given natural number, show that there exist only finitely many normal subgroups n of G with $[G:N]=n$.

Solution:

(a) Since H is finite, it is finitely generated and finitely presented, the latter of which follows from the fact that a finite group is fully determined by its Cayley table. We observe that any homomorphism $G \to H$ must map generators of G to generators of H and satisfy the relations of both G and H; since there are finitely many generators and relations for H, it follows that there are finitely many such homomorphisms.

(b) We observe that any normal subgroup N of G arises as the kernel of the homomorphism $G \to G/N$, so in particular, $[G:N]=n$ if and only if N is the kernel of a homomorphism from G to a group of order n. There are finitely many such groups since any group of order n must embed in S_n by Cayley's theorem, and there are finitely many subgroups of S_n of any order, so by part (a), it follows that there are finitely many group homomorphisms from G to any group of order n, so there are finitely many normal subgroups N with $[G:N]=n$.

Ring Theory

Ring Theory Notes

Localization

If R is a (commutative, unital) ring, then we may consider a subset $S \subseteq R$ such that $1 \in S$, $0 \notin S$, and for any $s_1, s_2 \in S$, we have $s_1 s_2 \in S$ (such an S is called a *multiplicative set*). Then, we may construct the *ring of fractions* $S^{-1}R$ as the set of equivalence classes in $R \times S$ with $(r_1, s_1) \sim (r_2, s_2)$ if and only if $r_1 s_2 - r_2 s_1 = 0$. We write $[(r,s)] = \frac{r}{s}$, and define addition and multiplication by

$$\begin{aligned}\frac{r_1}{s_1} + \frac{r_2}{s_2} &= \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}\\ \frac{r_1}{s_1}\frac{r_2}{s_2} &= \frac{r_1 r_2}{s_1 s_2}.\end{aligned}$$

There is a natural embedding ι of R into $S^{-1}R$ by taking $r \mapsto \frac{r}{1}$.

Theorem: Let R be a commutative ring with unity, S a multiplicatively closed subset, and $\phi\colon R \to T$ a ring homomorphism that maps S to units in T. Then, there is a unique homomorphism $\varphi\colon S^{-1}R \to T$ such that $\varphi \circ \iota = \phi$.

$$\begin{tikzcd} R \arrow[r, "\iota"] \arrow[dr, "\phi"'] & S^{-1}R \arrow[d, "\varphi"] \\ & T \end{tikzcd}$$

Proof. Define

$$\varphi\left(\frac{r}{s}\right) = \phi(r)\phi(s)^{-1}.$$

□

Primality in the Gaussian Integers

We will discuss primality in the Gaussian integers. Recall that the Gaussian integers are given by

$$\begin{aligned}\mathbb{Z}[i] &= \{a + bi \mid a, b \in \mathbb{Z}\}\\ &\cong \frac{\mathbb{Z}[x]}{(x^2+1)}.\end{aligned}$$

The Gaussian integers are equipped with a norm $N: \mathbb{Z}[i] \to \mathbb{Z}_{\geq 0}$ given by $N(a+bi) = (a+bi)(a-bi) = a^2+b^2$, which is a multiplicative norm satisfying $N(zw) = N(z)N(w)$. Furthermore, with this norm, the Gaussian integers form a Euclidean domain. The units of $\mathbb{Z}[i]$ are ± 1 and $\pm i$.

Lemma: Let $q \in \mathbb{Z}[i]$. Then, there is a prime integer $p \in \mathbb{Z}$ such that $N(q) = p$ or $N(q) = p^2$.

Proof. Since q is not a unit, $N(q) \neq 1$. Therefore, $N(q)$ is a nontrivial product of prime integers, and since q is prime in $\mathbb{Z}[i]$, q must divide one of the prime integer factors of $N(q)$. Let p be this prime integer.

Then, $q|p$ in $\mathbb{Z}[i]$, so $N(q)|N(p) = p^2$, so $N(q) = p$ or $N(q) = p^2$. □

Example: We observe that 3 is a prime element of $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a Euclidean domain, it is a UFD, so irreducibles are prime, meaning we only need to show that 3 is irreducible in $\mathbb{Z}[i]$.

Since $N(3) = 9$, it follows that the norm of any factor of 3 must be a divisor of 9. Therefore, it follows that the norm of any divisor of 3 is equal to 1, 3, or 9. If the norm is 9, then the factor is an associate of 3, while if the norm is 1, then the factor is a unit, so it follows that any nontrivial factor of 3 would have norm equal to 3. Yet, there are no such elements.

Meanwhile, 5 is not a prime element of $\mathbb{Z}[i]$, since we may write $5 = (2+i)(2-i)$.

We will now focus on proving one of the most celebrated theorems in number theory: Fermat's Theorem on sums of squares. We will say a prime integer $p \in \mathbb{Z}$ *splits* in $\mathbb{Z}[i]$ if it is not a prime element of $\mathbb{Z}[i]$. For example, 5 splits in $\mathbb{Z}[i]$ while 3 does not.

Lemma: A positive prime integer $p \in \mathbb{Z}$ splits in $\mathbb{Z}[i]$ if and only if it is the sum of two squares in $\mathbb{Z}$.

Proof. Suppose $p = a^2 + b^2$ with $a, b \in \mathbb{Z}$. Then,

$$p = (a+bi)(a-bi)$$

in $\mathbb{Z}[i]$, so since $N(a \pm bi) = a^2 + b^2 = p \neq 1$, it follows that neither of the factors is a unit in $\mathbb{Z}[i]$, so that p is not irreducible, hence not prime.

Conversely, suppose p is not irreducible in $\mathbb{Z}[i]$. Then, p has an irreducible factor $q \in \mathbb{Z}[i]$ that is not an associate of p. Since $q|p$, multiplicativity of the norm gives that $N(q)|N(p) = p^2$, hence $N(q) = p$ as q and p are not associates and q is not a unit. If we have $q = a + bi$, then

$$\begin{aligned}p &= N(q)\\ &= a^2 + b^2.\end{aligned}$$

□

Next, we discuss a characterization of splitting via a single congruence.

Lemma: A positive odd prime integer p splits in $\mathbb{Z}[i]$ if and only if it is congruent to 1 modulo 4.

Proof. Our goal is to understand when $\mathbb{Z}[i]/(p)$ is an integral domain. Yet, we have the series of isomorphisms

$$\begin{aligned}\frac{\mathbb{Z}[i]}{(p)} &\cong \frac{\mathbb{Z}[x]/\left(x^2+1\right)}{(p)}\\ &\cong \frac{\mathbb{Z}[x]}{(p, x^2+1)}\\ &\cong \frac{\mathbb{Z}[x]/(p)}{(x^2+1)}\end{aligned}$$

$$\cong \frac{\mathbb{Z}/p\mathbb{Z}[x]}{(x^2+1)}.$$

In particular, this means that p splits in $\mathbb{Z}[i]$ if and only if x^2+1 has a root in $\mathbb{Z}/p\mathbb{Z}$, or that there is some n such that $n^2 \equiv -1$ modulo p. We will show that this condition is equivalent to p being congruent to 1 modulo 4.

Let $G = (\mathbb{Z}/p\mathbb{Z})^{\times}$, and fix a generator g. Observe that p is odd, so $p-1 = 2\ell$ for some integer ℓ. Let $\overline{n}$ denote the class of an integer n mod (p). For any $n \notin (p)$, it follows that there is some m such that $\overline{n} = g^m$.

Observe that the class $\overline{-1}$ generates the unique subgroup of order 2 of G, and since g^{ℓ} does the same, it follows that $g^{\ell} = -1$. In particular, we have that $n^2 = -1$ modulo p if and only if $g^{2m} = g^{\ell}$, or that $2m \equiv \ell$ mod 2ℓ, or that $(p-1) = 2\ell$ is a multiple of 4, or $p \equiv 1$ mod 4. □

This gives the following.

Theorem (Fermat's Theorem on Sums of Squares)**:** A positive odd prime $p \in \mathbb{Z}$ is a sum of two squares if and only if $p \equiv 1$ mod 4.

Ring Theory Worked Solutions

Problem (August '05, Problem 3)**:** Let R be a commutative ring with 1 such that any ideal in R is principal. Let $S \subseteq R$ be a multiplicatively closed subset of R. Show that any ideal in the ring of fractions $S^{-1}R$ is also principal.

Solution: We claim that any ideal in $S^{-1}R$ is of the form $S^{-1}J$ for some ideal J of R. For this, consider a proper ideal I (one may choose their favorite maximal ideal) of $S^{-1}R$, and consider the ideal $J = I \cap R$. We claim that $J \cap S = \emptyset$.

Suppose not, and let $u \in J \cap S$, so by definition of the copy of R in $S^{-1}R$, we have $\frac{u}{1} \in J \cap S$. Since $J \cap S$ is an ideal, it follows that $\frac{u}{1}\frac{1}{u} \in J \cap S$, so $1 \in J \cap S$, so $1 \in J$, meaning $1 \in I$, meaning I is not proper, which is a contradiction.

Since $J \cap S = \emptyset$, it follows that $S^{-1}J$ is a well-defined set. We start by showing that $I \subseteq S^{-1}J$. For this, consider an element $\frac{a}{b} \in I$. Then, we have $\frac{a}{1} = \frac{a}{b}\frac{b}{1} \in I$, so $\frac{a}{b} \in S^{-1}J$. Additionally, if we have some element $\frac{a}{b} \in S^{-1}J$, then we have $\frac{a}{b} = \frac{1}{b}\left(\frac{a}{1}\right)$, where by definition, $\frac{a}{1} \in I$ as $\frac{a}{1} \in I \cap R$, so that $\frac{a}{b} \in I$ since I absorbs multiplication.

In the special case where every ideal of R is principal, it follows that every ideal of $S^{-1}R$ is of the form $S^{-1}I$, meaning that every ideal of $S^{-1}R$ is principal generated by $\left(\frac{a}{1}\right)$, where $I = (a)$.

Problem (August '20, Problem 4)**:** Let R be a commutative ring with 1. An ideal I of R is called irreducible if I cannot be written as $I = J \cap K$, where J and K are both ideals strictly containing I.

(a) Prove that every prime ideal is irreducible.

(b) Assume that R contains a nonzero nilpotent element. Prove that R contains an irreducible ideal which is not prime.

Solution:

(a) Let P be a prime ideal of R. Suppose toward contradiction that there exist proper ideals J and K with $P = J \cap K$ and $P \subsetneq J, K$. Choose $a \in J \setminus P$ and $b \in K \setminus P$. Then, $ab \in P$ since $ab \in J \cap K$ by absorption of multiplication. Yet, by the definition of prime ideal, it follows that either $a \in P$ or $b \in P$, which contradicts the definition of a and b.

(b) Suppose $a \in R$ is a nonzero nilpotent element. We consider the set $\mathcal{S}$ of ideals that do not contain a, ordered by inclusion. This set is nonzero since the zero ideal is in this set and a is not zero. If $\mathcal{C}$ is a chain in $\mathcal{S}$, then the union of the ideals in $\mathcal{C}$ is an upper bound in $\mathcal{S}$. If it were not the case that the union was in $\mathcal{S}$, then we would have some ideal $I \in \mathcal{C}$ with $a \in I$, which would imply that $I \notin \mathcal{S}$.

Therefore, by Zorn's Lemma, there is a maximal element M of $\mathcal{S}$. Observe that the nilradical of R is the intersection of all the prime ideals, so since M does not contain a, it follows that M is not prime.

Suppose toward contradiction that M is not irreducible. Then, there would be ideals J and K where $M \subsetneq J, K$ with $M = J \cap K$. Without loss of generality, if $a \notin J$, then it would follow that $J \in \mathcal{S}$, which would violate the maximality of M. Meanwhile, if $a \in J$ and $a \in K$, then we would have $a \in J \cap K = M$, which would violate the fact that $M \in \mathcal{S}$.

Problem (January '22, Problem 3): Let $f(x,y), g(x,y) \in \mathbb{C}[x,y]$ be two polynomials that do not have a non-constant common factor.

(i) Show that f and g are relatively prime as elements of $\mathbb{C}(x)[y]$ and $\mathbb{C}(y)[x]$, where $\mathbb{C}(x)$ and $\mathbb{C}(y)$ are the fields of rational functions (i.e., the fraction fields of $\mathbb{C}[x]$ and $\mathbb{C}[y]$ respectively). Give a detailed argument.

(ii) Show that the system of polynomial equations $f(x,y) = 0$ and $g(x,y) = 0$ has finitely many solutions in $\mathbb{C}^2$.

Solution:

(i) Considering $\mathbb{C}[x,y] \cong (\mathbb{C}[x])[y]$ as the ring of polynomials in y over the integral domain $\mathbb{C}[x]$, then since $\mathbb{C}$ is a field, $\mathbb{C}[x]$ is a PID, hence a UFD, so $\mathbb{C}[x,y]$ is a UFD. In particular, the factorizations of $f(x,y)$ and $g(x,y)$ into irreducible polynomials in y over the underlying domain $\mathbb{C}[x]$ is unique. By Gauss's Lemma, irreducible polynomials in y over $\mathbb{C}[x]$ remain irreducible over $\mathbb{C}(x)$, so $f(x,y)$ and $g(x,y)$ do not share any factors in $\mathbb{C}(x)[y]$, hence they are relatively prime. A similar argument applies to $\mathbb{C}(y)[x]$ by switching the roles of x and y.

(ii)

Field/Galois Theory

Field Theory Notes

Irreducibility of Cyclotomic Polynomials

We will let Φ_n denote the nth cyclotomic polynomial, defined by

$$\Phi_n(x) = \prod_{\gcd(d,n)=1} \left(x - \zeta_n^d\right).$$

Recall that the cyclotomic polynomials satisfy

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Theorem: If p does not divide n, then the irreducible factors of $\overline{\Phi_n(x)} \in \mathbb{F}_p[x]$ are distinct and each has degree equal to the order of p modulo n.

Proof. We observe that $\Phi_n(x) | (x^n - 1)$ in $\mathbb{Z}[x]$, so the divisibility relation is preserved when reducing modulo p, meaning that $\overline{\Phi_n(x)}$ is separable in $\mathbb{F}_p[x]$ as $x^n - \overline{1}$ is separable in $\mathbb{F}_p[x]$ by the assumption that $\gcd(p,n) = 1$.

Let α be a root of $\overline{\Phi_n(x)}$ in an extension of $\mathbb{F}_p$. We will show that α is a primitive nth root of unity. Since $\overline{\Phi_n(x)} \big| \left(x^n - \overline{1}\right)$, it follows that we have $\alpha^n = 1$. If α were not of order n, then it has some order m that properly divides n. In particular, this means that α is a root of some $\Phi_d(x)$ with $d|n$ properly. In particular, since $d|n$, it follows that $\overline{\Phi_n(x)\Phi_d(x)}$ divides $x^n - \overline{1}$, meaning that α is a double root of $x^n - \overline{1}$, contradicting the fact that $x^n - \overline{1}$ does not have any repeated roots. Therefore, α is a primitive root of unity.

If $\pi(x)$ is an irreducible factor of $\overline{\Phi_n(x)}$ in $\mathbb{F}_p[x]$, and α is a root of $\pi(x)$, then α is a primitive nth root of unity, so the degree of π, which is equal to $\left[\mathbb{F}_p(\alpha) : \mathbb{F}_p\right]$, is the order of p modulo n. $\square$

Field Theory Worked Solutions

Problem (September '84, Problem 7): Factor $x^8 - 1$ into irreducibles in $\mathbb{Z}/p\mathbb{Z}[x]$ for all primes p.

Solution: We start by understanding the case where $p = 2$. Then, we observe that $x^8 - 1 = x^{2^3} - 1$, so that $x^8 - 1 = (x-1)^{2^3}$.

We start by observing that over $\mathbb{Z}$, $x^8 - 1$ factors into irreducibles given by

$$x^8 - 1 = \Phi_1(x)\Phi_2(x)\Phi_4(x)\Phi_8(x),$$

where $\Phi_n(x)$ denotes the nth cyclotomic polynomial. In particular, we observe that these cyclotomic polynomials are given by

$$\begin{aligned}\Phi_1(x) &= x - 1\\ \Phi_2(x) &= x + 1\\ \Phi_4(x) &= x^2 + 1\\ \Phi_8(x) &= x^4 + 1.\end{aligned}$$

Therefore, we must understand the factorization into irreducibles over $\mathbb{Z}/p\mathbb{Z}$ of $x^2 + 1$ and $x^4 + 1$. When p is congruent to 1 modulo 4, then -1 is a square modulo p, so we may factor $x^2 + 1 = (x-a)(x+a)$. When p is not congruent to 1 modulo 4, then -1 is not a square modulo p, meaning that $x^2 + 1$ is irreducible.

Now, we observe that the square of every odd number is congruent to 1 modulo 8, so that $x^8 - 1$ divides $x^{p^2-1} - 1$, so we have that $x^4 + 1$ factors completely over $\mathbb{F}_{p^2}$. In particular, we may factor $x^4 + 1$ into the product of the minimal polynomials of the roots of $x^4 + 1$ in $\mathbb{F}_{p^2}$ over $\mathbb{F}_p$.

Problem (August '99, Problem 9): If p is a prime number congruent to 1 mod 8, show that 2 is a quadratic residue mod p; i.e., there is an integer a such that $a^2 \equiv 2$ modulo p.

Solution: First, we show that there exists some α such that $\alpha^4 = -1$. For this, it suffices to show that the polynomial $x^4 + 1$ is not irreducible over $\mathbb{F}_p$. Since $\mathbb{F}_p$ is equal to the roots of the polynomial $p(x) = x^p - x$, we observe then that since $8|p-1$, it follows that $x^8 - 1$ divides $x^p - x = x\left(x^{p-1} - 1\right)$. Therefore, we have $x^4 + 1|p(x)$. Since $p(x)$ splits over $\mathbb{F}_p$, it follows that $x^4 + 1$ splits over $\mathbb{F}_p$, so that $x^4 + 1$ has a root in $\mathbb{F}_p$.

Consider now the element $a = \alpha + \alpha^{-1}$. Then, we have

$$\begin{aligned}a^2 &= \left(\alpha + \alpha^{-1}\right)^2\\ &= \alpha^2 + \alpha^{-2} + 2\\ &= 2\end{aligned}$$

mod p, since $\alpha^{-2} = -\alpha^2$.

Problem (August '21, Problem 7):

(a) Let $n \in \mathbb{N}$, and let F be an algebraically closed field of characteristic 0 or of characteristic $p > 0$ where $p \nmid n$. Prove that F contains a primitive nth root of unity.

(b) According to (a), $\overline{\mathbb{F}_3}$, the algebraic closure of the field of three elements contains a 13th root of unity. Call it ω. Compute $[\mathbb{F}_3(\omega) : \mathbb{F}_3]$.

Solution:

(a) If the characteristic of F equals zero, the smallest subfield of F containing 1 is isomorphic to $\mathbb{Q}$, so we will assume without loss of generality that $\mathbb{Q} \subseteq F$. The nth cyclotomic polynomial, $\Phi_n(x)$, then has $\operatorname{spl}_{\mathbb{Q}}(\Phi_n(x)) \subseteq F$ is a separable extension that contains a primitive nth root of unity. Therefore, F has a primitive nth root of unity.

If the characteristic of F equals p, then we observe that the polynomial $p(x) = x^n - 1$ over $\mathbb{F}_p$ has $p'(x) = nx^{n-1} \neq 0$, whence $p(x)$ splits fully into linear factors in the algebraic closure F as $p(x)$ is then separable. There is then some extension $\mathbb{F}_{p^d}$ of $\mathbb{F}_p$ containing all the roots of $p(x)$, and in particular, we have $n | d-1$. Since any element of $\left(\mathbb{F}_{p^d}\right)^\times$ is a root of unity, it follows that there is some primitive $(d-1)$th root of unity contained in $\mathbb{F}_{p^d}$, whence there is also a primitive nth root of unity.

(b) If ω is a primitive 13th root of unity, then ω is a root of the polynomial $p(x) = x^{13} - 1$. We observe that over $\mathbb{Z}$, hence over $\mathbb{F}_3$, we have that $p(x) | x^{26} - 1$, whence we have $p(x)$ divides $q(x) = x^{27} - x$. In particular, this means that $p(x)$ splits completely in $\mathbb{F}_{p^3} \subseteq F$ and does not split over $q_2(x) = x^9 - x$, so that $[\mathbb{F}_3(\omega) : \mathbb{F}_3] = 3$.

Problem (August '99, Problem 8)**:** If F is finite, show that every element $\alpha \in F$ is the sum $\alpha = \beta_1^2 + \beta_2^2$.

Solution: Since F is finite, we have that $F = \mathbb{F}_{p^n}$ for some prime p and some $n \geq 1$.

If $p = 2$, then the Frobenius map is given by squaring, and since F is finite, the map is surjective, so for any $\alpha \in \mathbb{F}_{2^n}$, there is some $\beta \in \mathbb{F}_{2^n}$ such that $\alpha = \beta^2$.

If $p \neq 2$, we consider the sets

$$\begin{aligned} A &= \left\{\beta^2 \,\middle|\, \beta \in F\right\} \\ B &= \{\alpha - \zeta \mid \zeta \in A\}. \end{aligned}$$

Observe that $\left(\mathbb{F}_{p^n}\right)^\times$ is cyclic, so $|A| = \frac{p^n+1}{2}$, and similarly, we have $|B| = \frac{p^n+1}{2}$. Yet, this means that $A \cap B \neq \emptyset$, so that there are some $\beta_1^2, \beta_2^2 \in A$ such that $\alpha - \beta_1^2 = \beta_2^2$, whence $\alpha = \beta_1^2 + \beta_2^2$.

Problem (August '08, Problem 3)**:** Prove or disprove the following statements about irreducibility.

(a) If $\mathrm{Gal}(E/\mathbb{Q}) = S_n$ for E the splitting field over $\mathbb{Q}$ of a polynomial $f(x) \in \mathbb{Q}[x]$ of degree n, then $f(x)$ must be irreducible.

(b) If α is algebraic over $\mathbb{Q}$, then its minimal polynomial $\mu_{\alpha,\mathbb{Q}}(x)$ *must* be irreducible.

(c) If $A \in \mathrm{Mat}_n(\mathbb{R})$, then its minimal polynomial $\mu_A(x)$ *must* be irreducible.

Solution:

(a) To start, we have that $\mathrm{Gal}(E/\mathbb{Q})$ acts transitively on the roots of $f(x)$; since there are at most n roots for $f(x)$ in its splitting field, it follows that $\mathrm{Gal}(E/\mathbb{Q}) \hookrightarrow S_m$, where m is the number of distinct roots of $f(x)$ in E. Yet, since $\mathrm{Gal}(E/\mathbb{Q}) = S_n$, it follows that $f(x)$ has n distinct roots in E, whence it is separable.

Since $f(x)$ is a separable polynomial whose Galois group is S_n, its Galois group is a transitive subgroup of S_n, whence $f(x)$ is irreducible.

Problem (Algebra II HW 11, Problem 6)**:** Let p be a prime, $K = \mathbb{F}_p(t)$, the field of rational functions in one variable over $\mathbb{F}_p$. Let $\sigma\colon K \to K$ be the unique automorphism of K such that $\sigma(t) = t + 1$, and $G = \langle\sigma\rangle$. Observe that G is cyclic of order p. Let $F = K^G$. Find an explicit element s such that $F = \mathbb{F}_p(s)$, and prove your answer.

Solution: Consider the product given by

$$s = \prod_{i=0}^{p-1}(t+i),$$

so that

$$\sigma(s) = \prod_{i=0}^{p-1}(\sigma(t)+i)$$

$$= \prod_{i=0}^{p-1}(t+i+1)$$
$$= s$$

since σ has order p. Therefore, $s \in K^G$, so that $\mathbb{F}_p(s) \subseteq K^G$.

Consider now the tower of extensions given by $\mathbb{F}_p(s) \subseteq K^G \subseteq K$. We may write

$$K = \left(\mathbb{F}_p(s)\right)(t),$$

so we may compute

$$\left[K : \mathbb{F}_p(s)\right] = \left[\mathbb{F}_p(s)(t) : \mathbb{F}_p(s)\right]$$
$$\leq p,$$

where the bound appears from the fact that t is a root of the polynomial

$$\prod_{i=0}^{p-1}(x+i) - s \in \mathbb{F}_p(s)[x].$$

Meanwhile, by Artin's Lemma, we know that K/K^G is Galois with Galois group G, whence $\left[K : K^G\right] = |G| = p$. Hence, we have $\left[K^G : \mathbb{F}_p(s)\right] = 1$, so that $\mathbb{F}_p(s) = K^G$.

Problem (Algebra II HW 11, Problem 1)**:** Let p be a prime, n a positive integer, and let $\Phi_n(x) = x^{p^n} - x$ be an element of $\mathbb{F}_p[x]$. Prove that Φ_n is equal to the product of all the monic irreducible polynomials in $\mathbb{F}_p[x]$ whose degrees divide n, with each polynomial appearing with multiplicity one.

Solution: First, we observe that if $\overline{\mathbb{F}_p}$ is an algebraic closure, then the splitting field of $\Phi_n(x)$ is given by $\mathbb{F}_{p^n} \subseteq \overline{\mathbb{F}_p}$. We may write

$$\Phi_n(x) = \prod_{\alpha \in \mathbb{F}_{p^n}} (x - \alpha)$$

split in $\overline{\mathbb{F}_p}$. Then, any irreducible factor f_i of Φ_n will define a subfield of the splitting field of $\Phi_n(x)$, which is given by $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$ for some n. We have that $m|n$, so that $\deg(f_i)|n$ and that

$$f_i(x) = \prod_{\alpha \in \mathbb{F}_{p^m}} (x - \alpha)$$

fully splits in $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$. In particular, this means that $f_i(x)$ has multiplicity one in the product decomposition of $\Phi_n(x)$.

Similarly, if $f_m(x)$ is a monic irreducible polynomial in $\mathbb{F}_p[x]$ with degree $m|n$, then $f_m(x)$ defines a field extension $\mathbb{F}_p \subseteq F \subseteq \mathbb{F}_{p^n} \subseteq \overline{\mathbb{F}_p}$ with $\left[F : \mathbb{F}_p\right] = m$, whence $F = \mathbb{F}_{p^m}$. Since $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$, it follows that every root of f_m in $\overline{\mathbb{F}_p}$ is a root in $\Phi_n(x)$, whence $f_m(x)|\Phi_n(x)$, and $f_m(x)$ has multiplicity one by the earlier discussion.

Therefore,

$$\Phi_n(x) = \prod_{m|n} f_m(x),$$

where $f_m(x)$ is a monic irreducible factor of degree m for $\Phi_n(x)$ and each f_m appears with multiplicity one.

Problem (January '20, Problem 8)**:** Let K/F be a field extension. Suppose that $K = F(a, b)$ for some $a, b \in K$ such that $a^2 \in F$ and $b^2 \in F$.

(a) Prove that $[K : F] \leq 4$.

(b) Give an example (with full proof) where $[K : F] = 4$.

(c) Assume that $\operatorname{char}(F) \neq 2$. Prove that the extension K/F is Galois.

(d) Now, assume that F is finite. Prove that $[K : F] \leq 2$.

Solution:

(a) We have

$$[F(a,b) : F] = [F(a,b) : F(b)][F(b) : F].$$

Observe that $\mu_{a,F(b)} | \mu_{a,F}$, with $\mu_{a,F}$ and $\mu_{b,F}$ having degree 2. Therefore, we have

$$\begin{aligned}[F(a,b) : F] &= [F(a,b) : F(b)][F(b) : F] \\ &\leq 4.\end{aligned}$$

(b) We consider the case of $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$. We start by noting that 2 and 3 are coprime. Observe that if $a + b\sqrt{2} = \sqrt{3}$ with $a, b \in \mathbb{Q}$, then we get

$$3 = a^2 + 2b^2 + 2ab\sqrt{2},$$

giving that $\sqrt{2}$ is rational, which is a contradiction. Therefore, we have that $\sqrt{3}$ is not contained in $\mathbb{Q}\left(\sqrt{2}\right)$, whence the minimal polynomial is still degree 2. Therefore, we get

$$\begin{aligned}\left[\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right) : \mathbb{Q}\right] &= \left[\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right) : \mathbb{Q}\left(\sqrt{2}\right)\right]\left[\mathbb{Q}\left(\sqrt{2}\right) : \mathbb{Q}\right] \\ &= 4.\end{aligned}$$

(c) We observe that a and b are roots of the separable polynomials $x^2 - a^2$ and $x^2 - b^2$ respectively. We have that $F(a,b) = F(a)F(b)$ can be expressed as a compositum, and that $F(a)/F$ and $F(b)/F$ are separable extensions of degree 2, hence they are Galois extensions. Since the compositum of two Galois extensions is Galois, we get that K/F is Galois.

(d) If F is finite, then $F = \mathbb{F}_q$ for some $q = p^m$. Since $F(a)/F$ and $F(b)/F$ are field extensions of degree at most 2, the result follows if either $F(a) = F$ or $F(b) = F$. Else, if both $F(a)/F$ and $F(b)/F$ are extensions of degree 2, then by uniqueness of finite fields, they are both equal to the field $\mathbb{F}_{q^2}$, whence $[F(a,b) : F] \leq 2$.

Galois Theory Notes

General Galois Theory Notes

Theorem: If $f(x) \in K[x]$ is a separable polynomial of degree n, then f is irreducible if and only if the Galois group $\operatorname{Gal}(\operatorname{spl}_K(f(x))/K)$ is a transitive subgroup of S_n.

Proof. If $f(x)$ is irreducible, then we know that the Galois group acts transitively on the roots.

Meanwhile, if $f(x)$ is reducible, then $f(x)$ is a product of distinct irreducibles since it is separable. Let r_i and r_j be roots of different irreducible factors of $f(x)$. Then, for each $\sigma \in \operatorname{Gal}(\operatorname{spl}_K(f(x))/K)$, we have $\sigma(r_i)$ has the same minimal polynomial over K as r_i, so we cannot have $\sigma(r_i) = r_j$, whence the Galois group is not transitive. $\square$

Galois Groups of Polynomials

Definition: Let $x_1,\dots,x_n$ be indeterminates. The *elementary symmetric functions* $s_1,\dots,s_n$ are defined by

$$\begin{aligned} s_1 &= x_1 + x_2 + \cdots + x_n \\ s_2 &= \prod_{i<j} x_i x_j \\ &\vdots \\ s_n &= x_1 x_2 \cdots x_n. \end{aligned}$$

The *general polynomial* of degree n is the polynomial

$$g(x) = (x - x_1)(x - x - 2)\cdots(x - x_n)$$

with roots $x_1,\dots,x_n$.

Observe that the general polynomial can be written as

$$(x - x_1)(x - x_2)\cdots(x - x_n) = x^n - s_1 x^{n-1} + \cdots + (-1)^n s_n.$$

This means that if F is some field, then the extension $F(x_1,\dots,x_n)$ is a Galois extension of the field $F(s_1,\dots,s_n)$ since it is the splitting field of the general polynomial of degree n. Furthermore, $F(s_1,\dots,s_n)$ is contained in the fixed field of the action of S_n on $F(x_1,\dots,x_n)$, where $\sigma \cdot x_i = x_{\sigma(i)}$ for any $\sigma \in S_n$. Since the fixed field of S_n has index $n!$ in $F(x_1,\dots,x_n)$ by the fundamental theorem of Galois theory, it follows that we have

$$[F(x_1,\dots,x_n) : F(s_1,\dots,s_n)] \leq n!,$$

whence $F(s_1,\dots,s_n)$ is the full fixed field of $F(x_1,\dots,x_n)$.

Theorem: The general polynomial

$$x^n - s_1 x^{n-1} + s_2 x^{n-2} + \cdots + (-1)^n s_n$$

over the field $F(s_1,\dots,s_n)$ is separable with Galois group S_n.

Now, we will discuss the discriminant, which is an essential tool in understanding the Galois groups of polynomials, especially those of degrees two, three, and four.

Definition: The *discriminant* of $x_1,\dots,x_n$ is given by

$$\Delta = \prod_{i<j} \left(x_i - x_j\right)^2.$$

The discriminant of a polynomial is the discriminant of the roots of the polynomial.

We start by observing that the discriminant is a symmetric function in $x_1,\dots,x_n$, so it is an element of the field $K = F(s_1,\dots,s_n)$.

Recalling the definition of the alternating group A_n, we have that $\sigma \in S_n$ is in A_n if and only if σ fixes

$$\sqrt{\Delta} = \prod_{i<j} \left(x_i - x_j\right)$$

in $\mathbb{Z}[x_1,\dots,x_n]$. Notice then that if F has characteristic not equal to 2, then $\sqrt{\Delta}$ generates the fixed field of A_n, which is a quadratic extension of K.

If $f(x)$ is a monic degree n polynomial with roots $\alpha_1,\dots,\alpha_n$, then the discriminant is given by

$$\Delta = \prod_{i<j} \left(\alpha_i - \alpha_j\right)^2.$$

Observe that $\Delta \in F$ since Δ is fixed by the Galois group. Furthermore, $\Delta = 0$ if and only if $f(x)$ is separable, so if F is perfect (i.e., characteristic zero or finite), then $f(x)$ is reducible if and only if $\Delta = 0$ since every irreducible polynomial over a perfect field is separable. Additionally, observe that $\sqrt{\Delta} \in F$ if and only if the Galois group of $f(x)$ is contained in A_n.

Now, we start our investigation with the polynomials of degree 2. The discriminant of a polynomial $x^2 + ax + b$ is given by $\Delta = (\alpha - \beta)^2$, which can be written as

$$\begin{aligned}\Delta &= s_1^2 - 4s_2\\ &= a^2 - 4b.\end{aligned}$$

The polynomial is separable if and only if $a^2 - 4b \neq 0$, and the Galois group is a subgroup of $S_2 = \mathbb{Z}/2\mathbb{Z}$. This means that the Galois group is trivial if and only if $a^2 - 4b$ is a rational square.

Next, we consider the cubic polynomials. If we have

$$f(x) = x^3 + ax^2 + bx + c,$$

which upon the substitution $x = y - a/3$, yields a reduced cubic

$$g(y) = y^3 + py + q,$$

where

$$\begin{aligned}p &= \frac{1}{3}\left(3b - a^2\right)\\ q &= \frac{1}{27}\left(2a^3 - 9ab + 27c\right).\end{aligned}$$

Since the roots of the polynomials f and g differ by the constant $a/3$, and the formula for the discriminant involves the difference of these roots, it follows that the polynomials have the same discriminant.

Letting α, β, γ be the roots of the polynomial $g(y)$, then we have

$$\begin{aligned}g'(y) &= (y-\alpha)(y-\beta) + (y-\alpha)(y-\gamma) + (y-\beta)(y-\gamma)\\ g'(\alpha) &= (\alpha-\beta)(\alpha-\gamma)\\ g'(\beta) &= (\beta-\alpha)(\beta-\gamma)\\ g'(\gamma) &= (\gamma-\alpha)(\gamma-\beta).\end{aligned}$$

Therefore, we have

$$\begin{aligned}\Delta &= ((\alpha-\beta)(\alpha-\gamma)(\beta-\gamma))^2\\ &= -g'(\alpha)g'(\beta)g'(\gamma).\end{aligned}$$

Since $g'(y) = 3y^2 + p$, it follows that

$$\begin{aligned}-\Delta &= \left(3\alpha^2 + p\right)\left(3\beta^2 + p\right)\left(3\gamma^2 + p\right)\\ &= 27\alpha^2\beta^2\gamma^2 + 9p\left(\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2\right) + 3p^2\left(\alpha^2 + \beta^2 + \gamma^2\right) + p^3.\end{aligned}$$

These expressions are elementary symmetric functions of the roots, where for g, $s_1 = 0$, $s_2 = p$, and $s_3 = -q$, whence

$$\Delta = -4p^3 - 27q^2.$$

Therefore, expressing Δ in terms of a, b, c, we have

$$\Delta = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc.$$

This allows us to compute the Galois group as follows.

(i) If $f(x)$ is reducible, then either $f(x)$ splits into three linear factors or into a linear factor and an irreducible quadratic. In the first case, the Galois group is trivial and in the second case, the Galois group is of order 2.

(ii) If $f(x)$ is irreducible, then a root of $f(x)$ generates a degree 3 extension over F. Therefore, the degree of the splitting field over F is divisible by 3.

Since the Galois group is a subgroup of S_3, it follows that there are only two options, either A_3 or S_3. In particular, the Galois group is A_3 if and only if the discriminant Δ is a square.

If Δ is a square, then the splitting field of the irreducible cubic $f(x)$ is obtained by adjoining any root of $f(x)$ to F. If Δ is not the square of an element of F, then the splitting field of $f(x)$ is of degree 6 over F, given by $F\left(\theta, \sqrt{\Delta}\right)$ where θ is some root of $f(x)$. This extension is Galois over F generated by the automorphisms σ and τ, where σ takes θ to any other root of $f(x)$ fixing $\sqrt{\Delta}$ and τ takes $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ and fixing θ.

Finally, we discuss the Galois group of a quartic polynomial

$$f(x) = x^4 + ax^3 + bx^2 + cx + d.$$

Then, by substituting with $x = y - a/4$, this yields the quartic

$$g(y) = y^4 + py^2 + qy + r$$

with

$$\begin{aligned} p &= \frac{1}{8}\left(-3a^2 + 8b\right) \\ q &= \frac{1}{8}\left(a^3 - 4ab - 8c\right) \\ r &= \frac{1}{256}\left(-3a^4 + 16a^2b - 64ac + 256d\right). \end{aligned}$$

Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of $g(y)$ and G the Galois group for the splitting field of $g(y)$.

If $g(y)$ is reducible, then the following cases hold.

(i) If $g(y)$ splits into a linear and a cubic, then G is the Galois group of the cubic.

(ii) If $g(y)$ splits into two irreducible quadratics, then the splitting field is $F\left(\sqrt{\Delta_1}, \sqrt{\Delta_2}\right)$, where Δ_1 and Δ_2 are the discriminants of the quadratics.

If Δ_1 and Δ_2 do not differ by a square, then G is isomorphic to a Klein 4-subgroup of S_4.

If Δ_1 is a square multiplied by Δ_2, then $F\left(\sqrt{\Delta_1}, \sqrt{\Delta_2}\right) = F\left(\sqrt{\Delta_1}\right)$ is a quadratic extension with G isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

Now, we have the case of $g(y)$ irreducible.Then, the Galois group is transitive on its roots. There are five transitive subgroups (up to conjugation) of S_4, given by copies of S_4, A_4, D_4, V, C, where V denotes the Klein 4-group and C denotes the copy of $\mathbb{Z}/4\mathbb{Z}$.

We consider the elements

$$\begin{aligned} \theta_1 &= (\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4) \\ \theta_2 &= (\alpha_1 + \alpha_3)(\alpha_2 + \alpha_4) \\ \theta_3 &= (\alpha_1 + \alpha_4)(\alpha_2 + \alpha_3) \end{aligned}$$

inside the splitting field for $g(y)$. Notice that the elements are permuted among themselves by the permutations of S_4. The stabilizer of θ_1 is the dihedral group D_4, while the stabilizer of all three of θ_1, θ_2, and θ_3 is the copy of V.

The elementary symmetric functions in the θ_i are fixed by S_4, so they are in F. These elementary symmetric functions evaluated on $\theta_1, \dots, \theta_3$ are given by $2p$, $p^2 - 4r$, and $-q^2$. This gives that θ_1, θ_2, and θ_3 are the roots of

$$h(x) = x^3 - 2px^2 + \left(p^2 - 4r\right)x + q^2.$$

This is known as the *resolvent cubic* for the quartic $g(y)$. It can be shown (see DF p.614) that the discriminant Δ of the resolvent cubic is equal to the resolvent of $g(y)$. In particular, this gives

$$\Delta = 16p^4r - 4p^3q^2 - 128p^2r^2 + 144pq^2r - 27q^4 + 256r^3.$$

Remark: In the special case where $g(y) = y^4 + py^2 + r$ (which emerges during analysis of quartics whose roots are nested square roots), we have

$$\Delta = 16p^4r - 128p^2r^2 + 256r^3.$$

In particular, since the splitting field for the resolvent cubic is a subfield of the splitting field of the original quartic, the Galois group for $h(x)$ is a quotient of G, so knowing the action of the Galois group of $h(x)$ on the roots of $h(x)$ gives information about the Galois group of $g(y)$.

(i) If the resolvent cubic is irreducible, and Δ is not a square, then G is not contained in A_4 with the Galois group of the resolvent cubic equal to S_3. In particular, this means that the degree of the splitting field for $g(y)$ is divisible by 6, so that $G = S_4$.

(ii) If the resolvent cubic is irreducible, and Δ is a square, then G is a subgroup of A_4 and 3 divides the order of G since the Galois group of the resolvent cubic is A_3, so that $G = A_4$.

(iii) If the resolvent cubic is reducible, and $h(x)$ splits completely, then every element of G fixes the elements θ_1, θ_2, and θ_3, so $G = V$.

(iv) If the resolvent cubic splits into a linear factor and a quadratic, then G stabilizes θ_1 but not θ_2 or θ_3, so $G \subseteq D_4$ and $G \nsubseteq V$, so $G = D_4$ or $G = C$.

    Now, notice that $F\left(\sqrt{\Delta}\right)$ is the fixed field of the elements of $G \cap A_4$. We have $D_4 \cap A_4 = V$ and $C \cap A_4 = \{e, (1,3)(2,4)\}$, the latter of which is not transitive. Therefore, we have the case $D_4 \cap A_4 = V$ if and only if $g(y)$ is irreducible over $F\left(\sqrt{\Delta}\right)$.

Galois Theory Worked Solutions

Problem (August ‘89, Problem 8)**:** Let $\beta = \sqrt{2 + \sqrt{2}}$. Show $\mathbb{Q}(\beta)/\mathbb{Q}$ is Galois with G cyclic; set up the Galois correspondence.

Solution: Working over the algebraic numbers, we observe that $\sqrt{2+\sqrt{2}}$ is a root of the irreducible polynomial

$$f(x) = x^4 - 4x^2 + 2.$$

The discriminant is given by

$$\begin{aligned}\Delta &= 16(-4)^4(2) - 128(-4)^2(2)^2 + 256(2)^3\\ &= 2^{11},\end{aligned}$$

so that $\mathbb{Q}\left(\sqrt{\Delta}\right) = \mathbb{Q}\left(\sqrt{2}\right)$. Then, over $\mathbb{Q}\left(\sqrt{2}\right)$, we have

$$f(x) = \left(\left(x^2 - 2\right) - \sqrt{2}\right)\left(\left(x^2 - 2\right) + \sqrt{2}\right),$$

which means that $f(x)$ is not irreducible over $\mathbb{Q}\left(\sqrt{2}\right)$. In particular, this means that the Galois group is equal to $\mathbb{Z}/4\mathbb{Z}$, and in particular, since the degree of $\mathbb{Q}(\beta)$ over $\mathbb{Q}$ is 4, it follows that the extension is indeed Galois.

Problem (January '17, Problem 1)**:** Consider the polynomial $f(x) = x^4 - 2x^2 - 6$. Prove this polynomial is irreducible. Describe the splitting field of this polynomial over $\mathbb{Q}$, and the Galois group of this splitting field.

Solution: We observe that this polynomial is contained in $\mathbb{Z}[x]$, so viewing $f(x)$ as an element of $\mathbb{Z}[x]$, we have that $f(x)$ is monic and $6 \notin \left(2^2\right)$, where $(2) \subseteq \mathbb{Z}$ is the ideal generated by the prime 2. By Eisenstein's criterion, it follows that $f(x)$ is irreducible over $\mathbb{Z}$, and by Gauss's Lemma, $f(x)$ is irreducible over $\mathbb{Q}$. Let K denote the splitting field of $f(x)$ over $\mathbb{Q}$.

Considering the set of roots

$$W = \left\{\sqrt{1+\sqrt{7}}, -\sqrt{1+\sqrt{7}}, \sqrt{1-\sqrt{7}}, -\sqrt{1-\sqrt{7}}\right\},$$

we observe that if we let

$$L = \mathbb{Q}\left(\sqrt{1+\sqrt{7}}\right),$$

then $f(x)$ is reducible over L, but has only two roots in L, so that $[K : L] = 2$. Since $[L : \mathbb{Q}] = 4$, it follows that $[K : \mathbb{Q}] = 8$.

Since $f(x)$ is irreducible, it follows that the Galois group of $f(x)$ acts transitively on the roots of f, and has order 8, so that $G \cong D_8$.

Problem (August '02, Problem 9)**:** Construct a Galois extension of $\mathbb{Q}$ of degree 3.

Solution: Consider ζ_7, a primitive 7th root of unity over $\mathbb{Q}$. Then, we have that $\mathbb{Q}(\zeta_7)/\mathbb{Q}$ is a Galois extension, since $\mathbb{Q}(\zeta_7)$ is the splitting field of the separable polynomial $x^7 - 1$, and has $\operatorname{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong (\mathbb{Z}/7\mathbb{Z})^\times \cong \mathbb{Z}/6\mathbb{Z}$. A Galois extension of degree 3 corresponds to a subgroup of index 3 in $(\mathbb{Z}/7\mathbb{Z})^\times$, which is the unique subgroup of order 2 in $(\mathbb{Z}/7\mathbb{Z})^\times$. If we let $\langle\sigma\rangle \subseteq \operatorname{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ be this subgroup, then σ is given by an element of order 2 in $\operatorname{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$; since complex conjugation is a $\mathbb{Q}$-automorphism of $\mathbb{Q}(\zeta_7)$, it follows that σ is complex conjugation.

Observe now that

$$\sum_{i=0}^{1} \sigma^i(\zeta_7) = \zeta_7 + \zeta_7^{-1}$$

is invariant under σ, so that $\mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right) \subseteq \mathbb{Q}(\zeta_7)^\sigma$. Furthermore, letting $\alpha = \zeta_7 + \zeta_7^{-1}$, we have

$$\zeta_7^2 - \alpha\zeta_7 + 1 = 0,$$

whence $\mu_{\zeta_7, \mathbb{Q}(\alpha)}(x)$ has degree at most 2.

Therefore, we observe that $\mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right)$ is an extension with $\left[\mathbb{Q}(\zeta_7) : \mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right)\right] = 2$, whence $\mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right)$ is the desired extension of degree 3 over $\mathbb{Q}$.

Problem (August '08, Problem 8)**:**

(a) Find a splitting field E of the polynomial $x^4 + 3x^3 + 4x^2 + 3x + 3$ over $F = \mathbb{Q}$, and find its degree $[E : F]$.

(b) Find the Galois group $\operatorname{Gal}(E/F)$ of the extension field.

(c) Diagram the lattice of subgroups of the Galois group and the corresponding lattice of intermediate subfields of E/F.

Solution:

(a) Taking the factorization over $\mathbb{Q}$, we have that the polynomial factorizes as $\left(x^2 + 1\right)\left(x^2 + 3x + 3\right)$. Over

$\mathbb{C}$, we may factorize this as

$$p(x) = (x-i)(x+i)\left(x - \left(-\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)\right)\left(x - \left(-\frac{3}{2} - i\frac{\sqrt{3}}{2}\right)\right).$$

Adjoining the roots to $\mathbb{Q}$ gives that $E \subseteq \mathbb{Q}\left(i, \sqrt{3}\right) \subseteq E$. Therefore, $E = \mathbb{Q}\left(i, \sqrt{3}\right)$. We have that

$$\begin{aligned} [E : \mathbb{Q}] &= \left[\mathbb{Q}\left(i, \sqrt{3}\right) : \mathbb{Q}(i)\right][\mathbb{Q}(i) : \mathbb{Q}] \\ &= 4, \end{aligned}$$

where we use the fact that $\mu_{\sqrt{3},\mathbb{Q}(i)} = x^2 - 3$ is irreducible over $\mathbb{Q}(i)$ as 3 is not a square mod 4.

(b) We observe that the maps

$$\sigma = \begin{cases} i \mapsto -i \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$$

$$\tau = \begin{cases} i \mapsto i \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$$

are automorphisms of $\mathbb{Q}\left(i, \sqrt{3}\right)$ that determine the order 4 group $\operatorname{Gal}(E/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(c) The fixed field for the group $\langle\sigma\rangle$ is $\mathbb{Q}\left(\sqrt{3}\right)$, while the fixed field for the group $\langle\tau\rangle$ is $\mathbb{Q}(i)$.

Problem (August '18, Problem 7): What is the Galois group of $x^5 - 1$ over a field with 7 elements?

Solution: We seek to understand the factorization into irreducibles of $x^5 - 1$. Observe that we have

$$x^5 - 1 = (x-1)\left(x^4 + x^3 + x^2 + x + 1\right),$$

where we observe that the latter factor is equal to $\Phi_5(x)$, the 5th cyclotomic polynomial. Recalling the theorem that the nth cyclotomic polynomials factor over $\mathbb{F}_p$ into irreducibles whose degree is equal to the order of p modulo n, we observe that $7 \equiv 2 \bmod 5$, and 2 has order 4 in $(\mathbb{Z}/5\mathbb{Z})^\times$, so that $\Phi_5(x)$ is irreducible over $\mathbb{F}_7$.

In particular, this means that the Galois group of $x^5 - 1$ is equal to the Galois group of the splitting field; since $[\mathbb{F}_7(\alpha) : \mathbb{F}_7] = 4$, it follows that the Galois group is isomorphic to $\mathbb{Z}/4\mathbb{Z}$.

Problem (August '20, Problem 8): Let K be the splitting field of the polynomial $f(x) = \left(x^3 - 11\right)\left(x^2 + x - 1\right)$ over $\mathbb{Q}$.

(a) Find (with proof) the degree of the extension $K/\mathbb{Q}$.

(b) Determine the isomorphism class of the Galois group $\operatorname{Gal}(K/\mathbb{Q})$, and prove your answer.

Solution:

(a) Considering the embedding of $\mathbb{Q}[x] \subseteq \mathbb{C}[x]$, we observe that the roots of $f(x)$ are given by

$$A = \left\{\sqrt[3]{11}, \omega\sqrt[3]{11}, \omega^2\sqrt[3]{11}, \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}\right\},$$

where ω is a primitive cube root of unity. In particular, this means that the splitting field K of $f(x)$ is contained in $F = \mathbb{Q}\left(\omega, \sqrt[3]{11}, \sqrt{5}\right)$. Meanwhile, we have $F \subseteq K$ since we may find $\omega = \frac{\omega\sqrt[3]{11}}{\sqrt[3]{11}}$ and $\sqrt{5} = 1 + 2\frac{-1+\sqrt{5}}{2}$, so that $K = \mathbb{Q}\left(\omega, \sqrt[3]{11}, \sqrt{5}\right)$.

To compute the degree, we observe that the minimal polynomials are given by

$$\mu_{\sqrt[3]{11},\mathbb{Q}}(x) = x^3 - 11$$
$$\mu_{\omega,\mathbb{Q}}(x) = x^2 + x + 1$$
$$\mu_{\sqrt{5},\mathbb{Q}} = x^2 - 5.$$

Computing the degree, we observe that $\mu_{\sqrt[3]{11},\mathbb{Q}(\omega,\sqrt{5})} = \mu_{\sqrt[3]{11}}$, and similarly for $\mu_{\omega,\mathbb{Q}(\sqrt{5})}$ with $\mu_{\omega,\mathbb{Q}}$, so that the degree of $K/\mathbb{Q}$ is 12.

(b) First, we recall the theorem that a separable polynomial $f(x)$ is irreducible if and only if the action of the Galois group on the roots of $f(x)$ is transitive. Since f is separable (as we are working over $\mathbb{Q}$) and not irreducible, it follows that the Galois group of $f(x)$ is not transitive.

Now, we observe that $f(x)$ has two irreducible factors — namely, $p(x) = x^3 - 11$ (which is irreducible by the Eisenstein criterion) and $q(x) = x^2 + x - 1$, and in particular, the extension $K/\mathbb{Q}$ is equal to the compositum of the splitting fields of $p(x)$ and $q(x)$, which we label as L and M respectively. Since these are finite degree extensions whose intersection is equal to $\mathbb{Q}$, it follows that

$$\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{Gal}(L/\mathbb{Q}) \times \mathrm{Gal}(M/\mathbb{Q}).$$

We start by seeing that $L = \mathbb{Q}\left(\omega, \sqrt[3]{11}\right)$. Since $p(x)$ has degree 3 and not all of its roots are real, it follows that $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$, while since M has degree 2 over $\mathbb{Q}$, it follows that $\mathrm{Gal}(M/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}$, so that $\mathrm{Gal}(K/\mathbb{Q}) \cong S_3 \times \mathbb{Z}/2\mathbb{Z}$.

Problem (January '98, Problem 5): The Galois group G of a polynomial $f(x) \in F[x]$ of degree 4 is known to contain a subgroup isomorphic to the dihedral group D_4.

(a) Show that $f(x)$ is irreducible.

(b) If some root r_i of $f(x)$ lies in the subfield $F\left(r_j, r_k\right)$ generated by two other roots, show that $F\left(r_j, r_k\right)$ is a splitting field for $f(x)$ and that $G = D_4$.

Solution:

(a) We observe that the group S_4 is of order 24, and by the Sylow theorems, there are three conjugate subgroups of order 8, each of which is isomorphic to D_4. Since each of these copies of D_4 is transitive, it follows that the Galois group of $f(x)$ acts transitively on the roots of $f(x)$. Since the Galois group of $f(x)$ is transitive, it follows that $f(x)$ is irreducible.

(b)

Problem (Algebra II 2019 Midterm II, Problem 2): Let K/F be a finite Galois extension with $G = \mathrm{Gal}(K/F)$.

(a) Assume that G is a simple group, with $\alpha \in K \setminus F$ and $\mu_{\alpha,F}(x)$ the minimal polynomial of α over F. Prove that K is a splitting field for $\mu_{\alpha,F}(x)$.

(b) Let $n = [K : F]$, and fix integers m and ℓ with $m\ell = n$. Find a condition on the subfield lattice of K/F which is equivalent to the following: G can be written as a semidirect product $A \rtimes B$ for some subgroups A and B with $|A| = m$ and $|B| = \ell$.

Solution:

(a)

(b) By the definition of the semidirect product, we have that $A \trianglelefteq G$, $G = AB$, and $A \cap B = \{e\}$. In particular, since $|AB| = \frac{|A||B|}{|A\cap B|}$, we have that $G = AB$ is equivalent to $A \cap B = \{e\}$.

We have that $A \trianglelefteq G$ if and only if K^A/F is Galois, that $G = AB$ if and only if $K^{AB} = F$, and that $A \cap B = \{e\}$ if and only if $K^{A\cap B} = K$.

Observe now that $K^{AB} = K^A \cap K^B$. This follows from the fact that $\alpha \in K^{AB} = F$ if and only if $g(\alpha) = \alpha$

for any $g \in AB$, hence for all $g \in A$ and $g \in B$.

Therefore, we claim that $G = A \rtimes B$ with $|A| = m$ and $|B| = \ell$ if and only if K/F contains subfields M_A and M_B such that

(i) $[M_A : F] = \ell$, $[M_B : F] = m$;

(ii) M_A/F is Galois;

(iii) $M_A \cap M_B = F$.

The forward direction follows from our earlier work by setting $M_A = K^A$ and $M_B = K^B$.

Problem (DF Problem 14.2.12): Determine the Galois group of the splitting field over $\mathbb{Q}$ of $x^4 - 14x^2 + 9$.

Solution: Factorizing over $\mathbb{C}$, we observe that

$$\begin{aligned}
p(x) &= x^4 - 14x^2 + 9\\
&= \left(x^2 - 7\right)^2 - 40\\
&= \left(x - \sqrt{7 + 2\sqrt{10}}\right)\left(x + \sqrt{7 + 2\sqrt{10}}\right)\left(x - \sqrt{7 - 2\sqrt{10}}\right)\left(x + \sqrt{7 - 2\sqrt{10}}\right).
\end{aligned}$$

If we let

$$\begin{aligned}
\alpha &= \sqrt{7 + 2\sqrt{10}}\\
\beta &= -\sqrt{7 + 2\sqrt{10}}\\
\gamma &= \sqrt{7 - 2\sqrt{10}}\\
\delta &= -\sqrt{7 - 2\sqrt{10}},
\end{aligned}$$

then we observe that $\alpha\gamma = 3$, whence $\gamma \in \mathbb{Q}(\alpha)$ so that $\mathbb{Q}(\alpha)/\mathbb{Q}$ is the splitting field for $p(x)$. Since $p(x)$ is irreducible, it follows that $\mathbb{Q}(\alpha)/\mathbb{Q}$ has degree 4. Furthermore, since $p(x)$ is a separable irreducible polynomial, there are automorphisms taking α to every other root, giving the automorphism group

$$G = \{\mathrm{id}, (\alpha \mapsto \gamma), (\alpha \mapsto \beta), (\alpha \mapsto \delta)\}.$$

Observe that all of these automorphisms fix $\mathbb{Q}$ and have order 2, meaning that $G \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Problem (August '21, Problem 6): Consider $f(x) = x^4 + ax^2 - 1 \in \mathbb{Q}[x]$, where $a \in \mathbb{Z} \setminus \{0\}$. Let K be the splitting field of $f(x)$ over $\mathbb{Q}$.

(a) Show that $f(x)$ is irreducible.

(b) Show that $i \in K$.

(c) Determine the isomorphism class of the Galois group $\mathrm{Gal}(K/\mathbb{Q})$.

Solution:

(a) From the rational roots theorem, we know that any rational root of $f(x)$ must be either 1 or -1. Since both $f(1)$ and $f(-1)$ are equal to a, and $a \neq 0$, it follows that $f(x)$ has no rational roots, so $f(x)$ is irreducible.

(b) Since $f(x)$ is irreducible over K, it follows that $f(x)$ is separable, and upon completing the square we may write

$$x^2 + ax^2 - 1 = (x - \alpha)(x + \alpha)(x - \beta)(x + \beta)$$

for some $\alpha, \beta \in K$, meaning that we have $\alpha^2\beta^2 = -1$. Therefore, it follows that $\alpha\beta = i$, so $i \in K$.

(c) To determine the isomorphism class of the Galois group $\operatorname{Gal}(K/\mathbb{Q})$, we start by considering the three roots

$$\begin{aligned}\theta_1 &= 0\\ \theta_2 &= -(\alpha+\beta)^2\\ &= -\left(\alpha^2+\beta^2+2i\right)\\ \theta_3 &= -(\alpha-\beta)^2\\ &= -\left(\alpha^2+\beta^2-2i\right)\end{aligned}$$

Observe that $\theta_1 \in \mathbb{Q}$, meaning that the Galois group is isomorphic to either D_4 or $\mathbb{Z}/4\mathbb{Z}$ as θ_1 is stabilized by the Galois group. To distinguish, we will compute the order of $K = \mathbb{Q}(\alpha, i)$.

For this, it suffices to compute $[K : \mathbb{Q}(\alpha)]$. Since we may write

$$\begin{aligned}\alpha &= \sqrt{\frac{-a+\sqrt{1+a^2}}{2}}\\ &\in \mathbb{R},\end{aligned}$$

it follows that $\mathbb{Q}(\alpha) \subsetneq K$, so that

$$\begin{aligned}[K:\mathbb{Q}] &= [K:\mathbb{Q}(\alpha)][\mathbb{Q}(\alpha):\mathbb{Q}]\\ &= 8,\end{aligned}$$

where we use the fact that f is irreducible to determine that $[\mathbb{Q}(\alpha):\mathbb{Q}] = 4$.